

POTTSTOWN, PA, USA

WMO: 725109

Lat: 40.238N

Lon: 75.555W

Elev: 88

StdP: 100.27

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54782

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.3	-20.9	0.6	-11.3	-18.0	0.8	-8.5	9.5	-4.2	8.7	-2.9	2.8	300	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.0	23.5	31.6	22.7	30.2	22.2	24.9	30.4	24.2	28.8	23.5	27.9	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.4	27.0	22.8	17.7	26.2	22.2	17.1	25.8	76.0	30.4	73.0	29.1	70.4	28.0	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.0	-15.7	34.9	2.7	2.2	-17.6	36.5	-19.2	37.8	-20.8	39.0	-22.8	40.6		
(5)			-16.5	26.4	2.5	0.7	-18.3	26.9	-19.8	27.3	-21.2	27.7	-23.1	28.2		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.8	0.3	5.0	11.4	16.9	21.4	24.2	23.1	19.4	12.9	7.0	2.2
(7)		DBStd	9.77	5.62	5.22	5.33	4.97	4.46	3.46	2.75	2.80	3.57	4.62	4.71	4.87
(8)		HDD10.0	1204	337	274	171	38	2	0	0	0	23	113	246	
(9)		HDD18.3	2866	593	504	412	214	82	12	0	2	30	178	339	499
(10)		CDD10.0	1931	2	3	17	82	218	342	441	404	281	113	24	6
(11)		CDD18.3	552	0	0	0	8	39	104	183	148	61	9	0	0
(12)		CDH23.3	4897	0	0	6	107	373	929	1724	1231	466	61	1	0
(13)		CDH26.7	1631	0	0	0	29	106	291	671	402	124	9	0	0
(14)	Wind	WSAvg	2.2	2.7	2.8	2.9	2.7	2.0	1.8	1.6	1.4	1.6	1.9	2.2	2.5
(15)	Precipitation	PrecAvg	1178	87	65	107	94	92	108	114	104	116	105	86	103
(16)		PrecMax	1568	160	155	181	183	181	265	235	231	297	255	135	218
(17)		PrecMin	852	35	14	23	41	25	42	14	34	13	6	27	20
(18)		PrecStd	176	37	30	39	40	36	57	50	44	71	63	30	46
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	17.0	19.1	23.5	30.2	31.6	33.8	35.8	34.1	32.4	27.7	21.9	18.7	
(20)		0.4% MCWB	12.5	13.4	15.1	18.8	21.8	23.6	23.8	23.7	22.3	20.0	16.5	16.1	
(21)		2% DB	13.3	14.7	19.4	26.6	29.5	31.6	33.4	32.3	30.0	24.9	19.2	15.2	
(22)		2% MCWB	10.4	10.3	12.8	16.7	21.0	22.5	23.8	23.3	21.9	19.3	15.0	13.3	
(23)		5% DB	9.8	10.8	16.5	23.2	27.5	29.9	32.0	30.3	27.8	22.8	17.3	12.3	
(24)		5% MCWB	7.0	6.8	11.0	14.9	19.9	22.0	23.1	22.6	20.3	17.9	13.4	10.1	
(25)		10% DB	7.3	7.8	13.7	20.5	25.2	28.2	30.4	28.8	26.1	20.8	14.8	9.6	
(26)		10% MCWB	4.8	4.3	9.2	13.6	18.4	20.9	22.6	21.8	19.8	16.8	11.2	7.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	14.8	15.3	16.5	20.3	23.5	24.9	26.2	25.9	24.5	22.3	18.0	16.4	
(28)		0.4% MCDB	15.9	18.3	21.9	26.7	29.7	31.5	32.9	31.6	29.2	24.7	20.3	17.8	
(29)		2% WB	10.6	10.7	14.3	18.1	22.1	23.9	25.1	24.8	23.4	20.8	16.0	13.6	
(30)		2% MCDB	12.7	13.9	18.1	24.3	27.3	29.2	31.0	29.4	27.1	23.5	18.7	14.9	
(31)		5% WB	7.6	6.9	11.9	16.4	20.9	23.0	24.3	24.0	22.4	19.3	14.0	10.6	
(32)		5% MCDB	9.7	9.9	15.4	21.5	26.0	27.7	29.3	28.0	25.6	22.2	16.5	12.0	
(33)		10% WB	4.9	4.7	9.3	14.7	19.7	22.1	23.7	23.1	21.3	17.4	12.2	7.6	
(34)		10% MCDB	7.1	7.5	13.4	19.5	23.7	26.5	28.2	26.9	24.4	19.5	14.1	9.0	
(35)	Mean Daily Temperature Range	MDBR	8.3	9.1	10.2	11.5	11.1	10.5	10.4	10.1	10.8	10.3	9.9	8.2	
(36)		5% DB MCDBR	12.5	13.3	14.4	15.6	13.5	12.3	12.1	11.6	12.7	12.5	12.8	11.3	
(37)		5% WB MCWBR	9.9	9.8	9.4	7.9	6.4	5.3	4.6	4.8	5.9	7.0	8.8	9.7	
(38)		5% WB MCDBR	11.3	12.2	12.8	13.1	11.9	10.4	10.0	9.4	9.5	9.8	10.3	10.2	
(39)		5% WB MCWBR	9.7	9.9	9.8	8.0	6.2	5.2	4.2	4.4	5.4	6.4	8.4	10.1	
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.322	0.359	0.395	0.438	0.501	0.499	0.497	0.409	0.360	0.325	0.307	
(41)		taud	2.450	2.389	2.362	2.288	2.198	2.062	2.083	2.073	2.290	2.441	2.504	2.498	
(42)		Ebn at Noon	856	893	892	879	844	789	786	774	828	839	831	830	
(43)		Edh at Noon	79	97	111	128	143	165	160	157	119	92	75	71	
(44)	All-Sky Solar Radiation	RadAvg	1.86	2.76	3.65	4.75	5.30	5.79	5.93	5.16	4.26	3.02	2.13	1.60	
(45)		RadStd	0.16	0.31	0.28	0.44	0.64	0.47	0.29	0.32	0.52	0.32	0.19	0.17	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.54	N/A	N/A	N/A	N/A
(47)	Regional (72 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+123	+70	

Nomenclature: See separate page